

Mechanical properties of homogeneous and heterogeneous layered 2D materials

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Abstract—Transition metal dichalcogenides (TMDC) such as molybdenum disulfide (MoS_2) are 2D materials that are promising for flexible electronics and piezoelectric applications, but their low mechanical strength limits practical use. In this work, we study the mechanical properties of heterostructures containing MoS_2 and graphene, another 2D material with exceptional mechanical properties, using atomistic molecular dynamics simulations of nanoindentation. We consider bi- and tri-layer heterostructures where graphene either supports or encapsulates MoS_2 , and we compare to the monolayers and homogeneous bilayers. We extract mechanical properties (Young's modulus) from nanoindentation simulations. All of the heterostructures have larger Young's moduli than the mono- and bi-layer MoS_2 , demonstrating that graphene provides mechanical reinforcement regardless of layer stacking order. Our results demonstrate the potential of heterostructures to improve the mechanical properties of TMDC materials, which would increase their utility for device applications.

Keywords—mechanical properties; 2D materials; molecular dynamics simulation

I. INTRODUCTION

Two-dimensional (2D) materials such as graphene and molybdenum disulfide (MoS_2) are at the forefront of industrial and academic research, owing to their unique electronic, mechanical, thermal, and optical properties. Although graphene and MoS_2 possess a similar in-plane hexagonal lattice structure, their lattice structures out-of-plane are quite different. Graphene contains sp^2 -bonded carbon atoms, while MoS_2 constitutes a tetragonally aligned tri-layer between a transition metal (Mo) atom and two chalcogenide (S) atoms. As a result, the electronic and mechanical properties of graphene and MoS_2 are significantly different. Pristine graphene is metallic and mechanically stronger than steel, with a Young's modulus of ~ 1 TPa [1]. In contrast, MoS_2 is a semiconductor and is useful in device applications, but has Young's modulus of only about 0.25 TPa [2]. Recently, there have been efforts to combine these two materials for applications such as ultra-gain photodetectors [3], thermal interface materials [4], and superlubric materials [5]. Developing a deeper understanding of the electronic and mechanical properties of these heterostructures is critical for extending their applicability into other exotic applications such

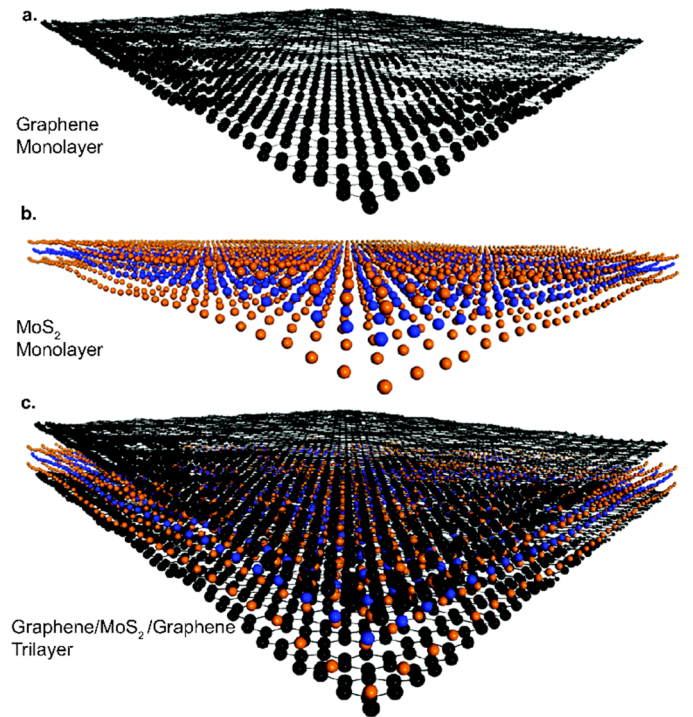


Fig. 1. Atomistic schematics of a) graphene (G) and b) MoS_2 (M) monolayers, and c) a graphene- MoS_2 -graphene (G/M/G) trilayer heterostructure. Carbon, molybdenum, and sulfur atoms are colored black, blue, and orange, respectively.

as excitonic, spintronics, and piezoelectric applications. In particular, piezoelectric applications demand a systematic study on how the mechanical properties depend on the layer stacking order. In this study, we use atomistic molecular dynamics simulations to study the mechanical deformation of graphene/ MoS_2 heterostructures with various stacking patterns relative to both single layers and homogeneous bilayers.

To mimic experimental methods for measuring the mechanical properties of 2D materials [1], we conduct simulated nanoindentation of several 2D homogeneous and heterogeneous systems. These systems are monolayers of either graphene (G) or MoS_2 (M), bi-layers of graphene (G/G) or MoS_2 (M/M), bi-

layer heterostructures with both graphene and MoS₂ (G/M and M/G), and tri-layer sandwich heterostructures (G/M/G and M/G/M).

II. MODELS AND METHODS

To construct single- and multi-layer models for the indentation study (Fig. 1), we obtained optimized structure parameters of the single layers from first-principle calculations performed using VASP tools [6, 7]. Using the Perdew-Burke-Ernzerhof (PBE) form of generalized-gradient approximation (GGA) [8, 9], optimized parameters for single layer graphene ($a = 2.46 \text{ \AA}$) and MoS₂ ($a = 3.179 \text{ \AA}$ and $c = 12.729 \text{ \AA}$) were obtained.

Using the DFT-generated lattice parameters, we constructed single layer, bilayer, and trilayer planar models as shown in the Fig. 1. We performed atomistic molecular dynamics (MD) simulations using LAMMPS (<http://lammps.sandia.gov>) [15]. The interatomic interactions for MoS₂ were described by a Stillinger-Weber potential, while carbon-carbon interactions in graphene were described by a bond-order potential with built-in long-range corrections (LCBOP) [16, 17]. The weak inter-layer van der Waals (vdW) interactions between MoS₂ layers were described by a Lennard-Jones (L-J) potential with energy and distance parameters $\epsilon = 6.93 \text{ meV}$ and $\sigma = 3.13 \text{ \AA}$ [16, 18]. In heterostructures, graphene and MoS₂ layers were coupled by vdW interactions, which were described using a L-J potential with $\epsilon = 3.95 \text{ meV}$ and $\sigma = 3.625 \text{ \AA}$, respectively [19]. The integration time step was 0.5 fs . L-J interactions were cut off at $10 \text{ \AA}$.

Prior to indentation simulations, all models were equilibrated using the following procedure. First, a small ($\sim 3 \text{ nm}^2$) material system was constructed with the surface normal to the z -dimension. The energy was minimized using steepest descent and conjugate gradient minimization, each for 10^4 steps, to remove highly unfavorable atomic interactions. The small system was then equilibrated during 100 picoseconds (ps) of constant temperature-constant pressure (NPT) MD at 300 K and 1 atm using the Nosé-Hoover thermostat and barostat with damping constants of 20 and 200 femtoseconds (fs), respectively [10]. The barostat acted independently on the x - and y -dimensions and the xy tilt factor. The small system was then replicated in the xy -plane to yield a large system of about $75 \times 75 \text{ nm}^2$, which was subsequently equilibrated for another 50 ps . Following this final equilibration, a circular section with radius $a = 30 \text{ nm}$ was identified in the center of the large system. Atoms within this radius were free to move during nanoindentation, while atoms outside this radius were clamped in place to form a rigid boundary. The 30 nm radius is comparable to previous simulation studies [11]. This simulation design mimics experimental methods for measuring the mechanical properties of 2D materials [1].

Following the above procedure, we obtained the force exerted on the 2D materials as a function of indentation depth (i.e., deflection, δ). A spherical indenter was assumed and described by a purely repulsive harmonic potential, $F(r) = -k(r-R)^2$, where F is the force on an atom at a distance r from the indenter with radius R (10 nm , similar to previous simulation and experimental studies) [1, 11], and k is the force constant ($217 \text{ meV/\AA}^3 \approx 5 \text{ kcal/mol/\AA}^3$). The force was zero for

$r > R$. Initially, we positioned the indenter in the center of the circular section with a vertical spacing of $\sim 2 \text{ \AA}$ between the indenter tip and the surface. The indenter was then lowered into the material at a speed of $0.5 \text{ \AA/ps}$, similar to previous simulation studies [11]. The force was then calculated as the total normal force on the spherical indenter. We define indentation depth (δ) as the difference between the initial position of the center-of-mass of the material and the lowest point of the indenter. During indentation, constant volume-constant temperature (NVT) MD was employed, where the temperature of the atoms inside the circular section was maintained at 300 K using the thermostat described above.

Intrinsic mechanical properties were then calculated by fitting the force (F)-deflection (δ) curves to the following relation [1, 12].

$$F(\delta) = c\delta + d\delta^3 \quad (1)$$

The thickness-normalized Young's modulus E^{2D} and pre-tension σ_0 of a membrane with thickness h can then be determined from the fitting coefficients (c, d) using the relations $c = \sigma_0 \pi h$ and $d = E^{2D} q^3 a^{-2} h$, where $q = (1.05 - 0.15\nu - 0.16\nu^2)^{-1}$ is a dimensionless constant, ν is the Poisson's ratio, and a is the hole radius. The Young's modulus of an equivalent bulk material, E , is calculated as $E^{2D} h^{-1}$ [13]. For monolayer graphene and MoS₂, we assume thicknesses of 0.335 nm and 0.609 nm , respectively [13], where the thickness of the multilayer structures are defined as the sum of the layer thicknesses. We use $\nu = 0.165$ for graphene, $\nu = 0.25$ for MoS₂, and the weighted average of these values for multilayer structures [14]. Though these values of ν are zeroth-order approximations, the dimensionless parameter q is relatively insensitive to ν . We only fit (1) to the low-deflection ($\delta \leq 10 \text{ nm}$) data [12].

III. RESULTS AND DISCUSSION

Force (F)-deflection (δ) curves obtained from nanoindentation simulations are shown in Fig. 2.

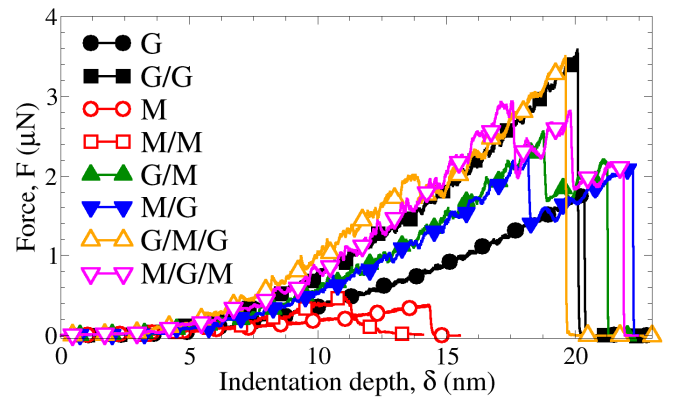


Fig. 2. Force (F)-deflection (δ) curves from simulated nanoindentation of 2D materials. G indicates a graphene layer, and M indicates a MoS₂ layer.

TABLE I. YOUNG'S MODULI OF 2D MONO-, BI-, AND TRI-LAYER SYSTEMS.

System	This study (TPa)	Literature values (TPa)
G	1.05	1.0 ± 0.1 [1]
G/G	1.06	1.04 ± 0.1 [20]
M	0.16	0.27 ± 0.1 [21]
M/M	0.27	0.2 ± 0.1 [21]
G/M	0.53	0.49 ± 0.05 [14]
M/G	0.52	--
G/M/G	0.68	0.56 [19]
M/G/M	0.45	--

The force increases slowly up to deflections of about 5 nm, at which point the force increases quickly up to failure. In the homogeneous bilayer systems, while these structures support larger forces than their monolayer counterparts, the MoS₂ bilayer is punctured more easily than the graphene bilayer. The bilayer heterostructures show similar maximum forces as the graphene monolayer, but exhibit two force peaks that correspond to the failure of MoS₂ and graphene, respectively. The trilayer heterostructures have maximum forces approaching the graphene bilayer, and exhibit two or more force peaks, corresponding to the rupture of individual layers, with MoS₂ layers always failing before graphene layers.

The Young's moduli for the 2D materials are shown in Table 1. For homogeneous structures, we have excellent agreement with experimental values: the moduli for graphene and MoS₂ mono- and bi-layers are ~ 1 TPa and ~ 0.25 TPa, respectively. The G/M and M/G bilayer heterostructures both have moduli of ~ 0.5 TPa, demonstrating that graphene provides mechanical reinforcement regardless of layer ordering. The trilayer heterostructures can provide larger or smaller Young's moduli than the bilayers, depending on the MoS₂ content. The moduli of heterostructures can be estimated accurately using the weighted average of the moduli of the layers [19].

These results suggest that MoS₂, supported or encapsulated by graphene to form bi- or tri-layer heterostructures, has superior mechanical properties compared with mono- and bi-layer MoS₂. Since other 2D materials, such as MoSe₂, WS₂, and WSe₂, have similar bonding and inter-layer vdW forces as MoS₂, our observations and approach open new avenues for the design of more robust 2D materials and for the characterization of their mechanical strength through large-scale atomistic modeling, which complements experimental efforts.

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